

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	CSE - AI	Date:	16/07/26

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A programmer wants to understand how an 8085 microprocessor executes different categories of instructions. To examine the execution sequence, write and execute an 8085 program containing data transfer, arithmetic, and branch instructions using an 8085 simulator. Observe the changes in the Program Counter (PC), Stack Pointer (SP), and memory locations during execution. Analyse how different instruction types progress through the fetch, decode, and execute cycle and explain how they influence the execution sequence.
2. A digital electronics engineer is verifying binary patterns used to control hardware devices. To investigate the correctness of the data, use the Windows Calculator (Programmer Mode) to perform bitwise operations (AND, OR, XOR, and NOT) on the given hexadecimal values. Analyse how the hexadecimal and binary representations change after each operation, and determine how bitwise operations are useful in computer systems. Support your analysis with suitable screenshots.
3. A user notices that a computer responds differently when opening a text editor, a media player, and a web browser simultaneously. To examine the execution of these tasks, perform the operations on the computer and observe the interaction between the CPU, memory, and I/O devices. Analyse which functional unit is primarily responsible for processing, storing, and transferring data during each operation, and explain how these components work together to complete the tasks.

Code of Conduct

I M. Koushik - 192372185 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Koushik Reddy

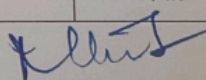
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

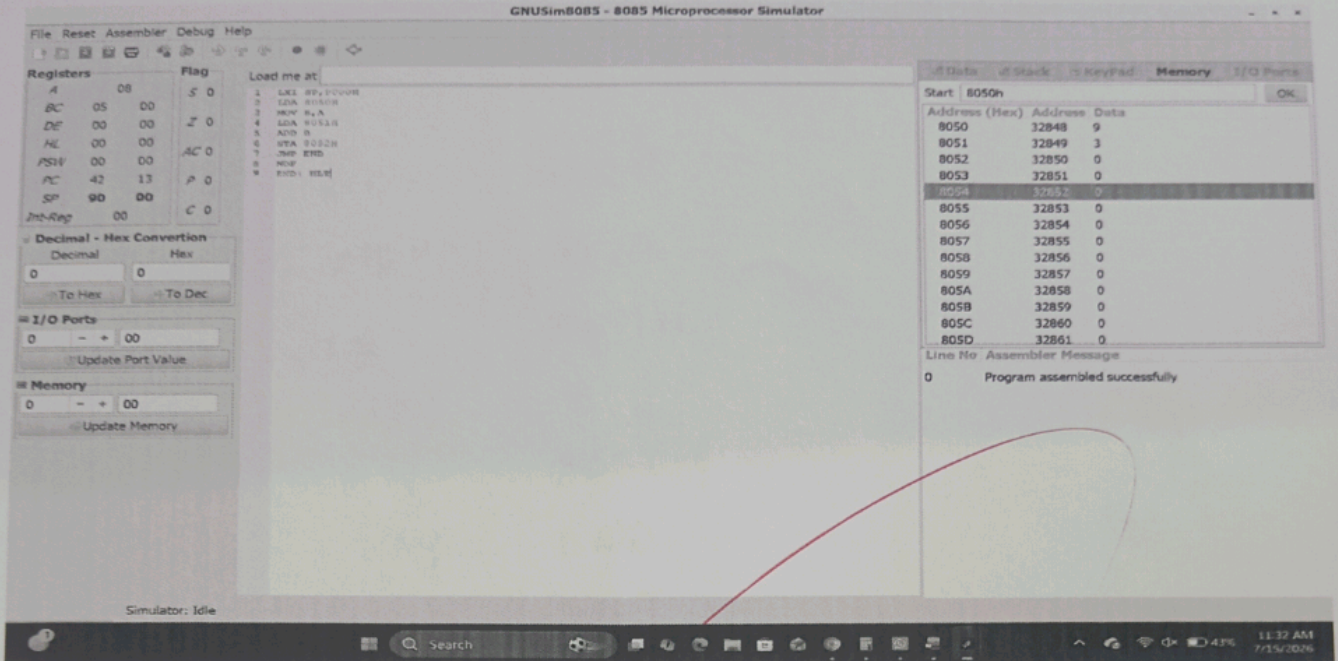
Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

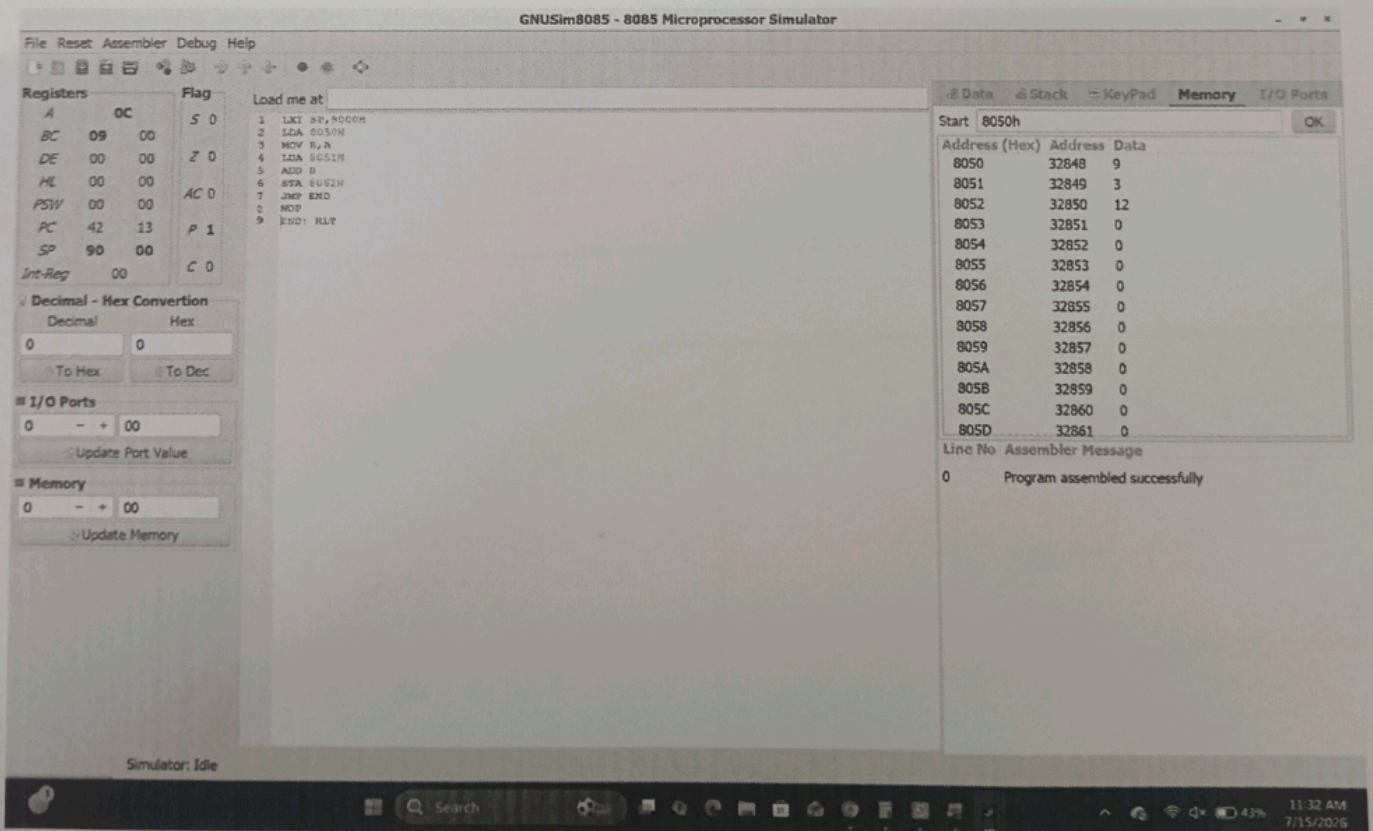
Course Coordinator


Faculty Incharge

BEFORE MEMORY



AFTER MEMORY



① Aim :- To study the execution of data transfer, arithmetic and branch instruction in 8085 microprocessor using an 8085 simulator.

Program :-

```
LDA 8050H
MOV B, A
LDA 8051H
ADD B
STA 8052H
JMP END
END : HLT
```

Execution cycle :-

Fetch → Decode → Execute.

observation Table :-

parameter	observation
Data 8050	09H
Data 8051	03H
Result 8052	09H 12H
program counter	changes after every instruction
Stack pointer	No change
Memory	8052 stores the sum.

②

Aim :- To perform bitwise operations (AND, OR, XOR, and NOT) on Hexadecimal values.

Procedure :-

→ Two Hexadecimal numbers

• A = 3CH → Binary :- 00111100

• B = A5H → A5 :- 10100101

Bitwise operations :-

operations	Results (Binary)	Result (Hex)
AND	00100100	24H
OR	10111101	BDH
XOR	10011001	99H
NOT (3C)	11000011	C3H

Result :- The bitwise operations were successfully performed using the windows calculator.

NOT Operator

Calculator - Programmer mode. The display shows the result of the NOT operation on the value 3C: **NOT(3C) = FFFF FFFF FFFF FFC3**. The left pane shows the value in different bases: HEX: FFFF FFFF FFFF FFC3, DEC: -61, OCT: 1 777 777 777 777 703, and BIN: 1111 1111 1111 1111 1111 1111 1111 1111. The right pane shows the selected QWORD format. The bottom pane shows the Bitwise and Bit shift menus.

XOR Operator

Calculator - Programmer mode. The display shows the result of the XOR operation on the values 3C and A5: **3C XOR A5 = 99**. The left pane shows the value in different bases: HEX: 99, DEC: 153, OCT: 231, and BIN: 1001 1001. The right pane shows the selected QWORD format. The bottom pane shows the Bitwise and Bit shift menus.

Calculator - Programmer mode. The display shows the result of the OR operation on the values 3C and A5: **3C OR A5 = BD**. The left pane shows the value in different bases: HEX: BD, DEC: 189, OCT: 275, and BIN: 1011 1101. The right pane shows the selected QWORD format. The bottom pane shows the Bitwise and Bit shift menus.

OR Operator

Calculator - Programmer mode. The display shows the result of the AND operation on the values 3C and A5: **3C AND A5 = 24**. The left pane shows the value in different bases: HEX: 24, DEC: 36, OCT: 44, and BIN: 0010 0100. The right pane shows the selected QWORD format. The bottom pane shows the Bitwise and Bit shift menus.

AND Operator

③

cpu, memory and I/O

open :-

• Notepad

• Media player

• google Chrome

component	Function
cpu	Executes Instructions & processes data.
RAM	stores active programs temporarily
I/O devices	keyboard, Mouse, monitor
System BUS	Transfers data b/w CPU, Memory and I/O devices.

Observation :-

- cpu usage increases while applications are running.
- RAM usage increases because multiple programs are loaded.
- I/O devices handle user input and display output.

Conclusion :-

cpu, memory and I/O devices work together to execute multiple applications efficiently.

MEDIA PLAYER- TASK MANAGER (PERFORMANCE)

